

Geometry-Informed Synthetic-to-Real Blade Damage Localization

Learning from simulated physical damage to identify defects in real blade photographs

Input photograph



Saved prediction



Reference damage



Actual real development photograph AEBIS_329: input, saved prediction (orange), and reference damage (teal). Individual-image IoU: 56.43%. [R, p. 47]

THE CENTRAL QUESTION

Can a model learn more useful damage cues from synthetic images when training also uses the known 3D damage location, then identify defects in real photographs using only the photograph?

Abstract

Main Jet will investigate label-efficient blade damage segmentation: marking the damaged pixels in a real image while limiting the number of real images that require defect annotations. The project combines a large existing synthetic image collection, a smaller set of synthetic images linked to known surfaces and physical changes, and public real inspection data. The key experiment will compare ordinary synthetic training against training on exactly the same images with additional geometric supervision.

Existing work establishes useful data-generation, localization and evaluation tools. It does not yet demonstrate that 3D supervision improves real-image segmentation. Earlier geometric losses improved some synthetic point-location errors while mask accuracy stayed flat or declined. Real-only training remains a strong baseline. These findings motivate a controlled study rather than an assumption that additional geometry must help. [R, pp. 23-30]

Proposed contribution

A reproducible blade-inspection study that isolates the value of geometric training information, measures performance at fixed real-label budgets, and documents when synthetic data helps or fails. A linked-data extension, verified labels and transparent evaluation protocol are intended outputs. Novelty and performance gains remain to be established.

SCOPE AND STATUS

Primary deliverable: real-image damage localization. Existing image-to-surface tools remain supporting assets; unknown 3D dent depth and removed volume are longer-term extensions. This document proposes a four-week pilot, subject to data readiness and measured run time. It launches no training and makes no publication-acceptance claim.

Audience: supervisor review, conference planning and funding discussions. Prepared from the supplied status report, with actual saved images, interactive evidence in the HTML companion, and references on the final page.

Research gap and intended contribution

The scientific question concerns the value of geometric supervision, beyond the value of synthetic images alone.

Why this problem matters

A useful inspection model must identify small defects despite changes in blade appearance, lighting, background and viewpoint. In this project, the available real collections have uneven labels and uncertain blade identities. Synthetic rendering provides controllable examples and exact labels, but good synthetic performance has not reliably transferred to the real images evaluated so far. [R, pp. 5-10, 23-24]

Related work	What it establishes	Implication for Main Jet
ZebraPose, WACV 2026 [1]	Bonetto and Ahmad train zebra detection and 2D pose models with synthetic data and evaluate on real imagery.	Adopt careful synthetic-to-real comparisons and real evaluation; a new architecture is not assumed necessary.
BladeSynth, 2025 [2]	An existing rendered blade dataset with defect masks and generation assets.	Synthetic blade images and defect generation are established. The project must add evidence beyond reusing them.
Looking 3D, CVPR 2024 [3]	Detects anomalies by comparing an image with a reference 3D shape.	Geometry-assisted inspection has prior work. Our primary deployment uses RGB without a supplied reference shape.
DefectSynth, 2026 [4]	The authors describe geometric synthetic data and training with geometry while retaining RGB-only prediction for concrete defects.	This is close conceptual prior art. Geometric training alone is not a sufficient novelty claim; the work is listed as under review.

Research questions

RQ1 - Synthetic transfer

At a fixed real-label budget, does synthetic pretraining improve real defect masks relative to a strong real-only baseline?

RQ2 - Added geometry

Does verified 3D supervision improve real masks beyond the same synthetic images with 2D labels, and beyond an ordinary 2D spatial loss?

RQ3 - Robustness and label efficiency

Do any benefits persist across eligible data sources, defect sizes, lighting conditions and reduced real-label budgets?

NOVELTY BOUNDARY

The intended contribution is a controlled, damage-aware training and evaluation study with linked image/surface evidence. No claim of being the first synthetic blade dataset, the first use of 3D supervision, or an already novel architecture is made. A broader literature comparison and convincing experimental differentiation are required before a strong conference claim.

Reference correction: the intended inspiration is the zebra-animal paper by Bonetto and Ahmad at WACV 2026 [1]. Page 66 of the supplied report discusses a different paper with the same ZebraPose name, concerning rigid-object pose estimation.

What data we actually have

These are inventory counts from the report, not the number of examples used in every experiment.

Source	Verified inventory and present role	Limitation for this proposal
Static BladeSynth archive	12,505 RGB files locally; 12,344 accepted: 2,500 healthy and 9,844 damaged. Existing split: 7,372 train, 1,793 development, 1,233 calibration, 1,946 test.	Broad 2D supervision. The local images have no verified per-image CAD/camera links for a 3D loss; the historical test is exposed.
Linked synthetic observations	582 accepted records; 571 distinct decoded RGB arrays. Includes nick, dent and search-reference/control observations.	A small, heterogeneous collection with limited source designs. Curate training eligibility; do not treat all records as independent defects.
AEBIS real masks	587 archive images; 528 validated damaged images used as 397 training and 131 development. Another 59 are reserved in the current protocol.	The 131 have been repeatedly used. Freshness of the 59 is uncertain because an earlier alternate archive was inspected. No clean images in the 528.
AeBAD-S and AeBAD-V	2,160 still-image records plus 3,410 video frames. Stills include damage masks; video annotations support frame-level analysis.	Official tests were already evaluated. Video frames are correlated. Labels and evaluation units differ.
Personal real photographs	174 original names, exported twice. 159 used: 116 outline-training and 43 development; 13 reserved and 2 excluded.	Existing labels describe blade outlines, not verified damage. Two exports do not create extra photographs.
Other intake	Lab archive: 493 RGB images with part polygons of unresolved defect meaning. RAD: 4,047 images from non-blade objects.	Neither is an additional qualified blade-defect mask training pool as currently documented.

Source: [R, pp. 5-10]. The published BladeSynth release describes 12,500 images [2]; the report records 12,505 local RGB files before filtering. These counts describe different inventories and should not be silently substituted.

Readiness work before model comparisons

Create one manifest recording image identity, source, label type, healthy/damaged status, physical blade or capture group when known, prior use, and split. Check exact duplicates and visually similar exports. Record which linked images have valid camera and surface associations. Keep all observations of the same known blade or capture sequence in the same train/development/test partition.

THE IMMEDIATE BOTTLENECK

The project has enough material for a controlled pilot. The bottleneck is suitable labels, independent examples and trustworthy evaluation. More renders of the same blade do not establish transfer to unfamiliar blade designs. Personal photographs need defect annotation and independent review before they can support damage-accuracy claims.

What the completed work tells us

The following are reported observations. They are not expected results of the proposed study.

Completed experiment	Reported result	Consequence for the proposal
Broad synthetic-to-real segmentation	Synthetic IoU 74.4%, but 3.979% on the evaluated real AeBAD-S images. [R, p. 23]	Synthetic success alone is insufficient. Real evaluation must be the main endpoint.
Early real-data adaptation	AEBIS IoU: real training from scratch 4.62 +/- 0.46%; synthetic initialization then real 16.02 +/- 1.86%. A larger pretrained model reached 29.62 +/- 4.29%. [R, p. 24]	Synthetic initialization showed potential in one setting. The larger-model comparison also changes the model and is not an isolated synthetic-data effect.
Later retained real model	IoU 61.941 +/- 3.081%, precision 81.963 +/- 3.969%, recall 71.083 +/- 1.453%; 397 train / 131 reused development images. [R, p. 24]	This is the strong historical reference. It used real task-training images and requires a fair rerun under the new protocol.
3D supervision for synthetic nicks	84 training images. Surface point cost improved 0.148299 to 0.061847; mask IoU fell 46.44% to 45.79%. [R, p. 26]	Better 3D point placement does not establish better segmentation. Retain a matched 2D spatial control.
Dent supervision	126 total RGB records; 42 training images from only three deformed training shapes. Inward IoU 2.68% to 3.02%; outward 9.09% to 8.97%. [R, pp. 8, 30]	The recent experiment is a small mechanism test, not full-dataset training or broad design generalization.
Learned continuous 3D displacement	On two exposed development shapes, changed-region error improved by about 11% and 26%, but whole-airfoil error was about 2.8x and 3.0x the unchanged-shape baseline. [R, pp. 102-103]	Continuous prediction exists, but false surface changes and alignment sensitivity prevent a reliable reconstruction claim.

Work retained and reused

Reuse the data filters, renderers, healthy/damaged meshes, masks, cameras, checked surface links, checkpoints, comparison infrastructure and visual prototype. Use the physical-change experiments to define failure controls and to motivate careful label meanings. Their development examples remain useful diagnostics.

EVIDENCE BOUNDARY

IoU is mask overlap: higher is better. Point distance and surface error measure different outcomes: lower is better. The +/- values above describe reported seed variation, not uncertainty across new physical blades. These report results were not independently rerun while preparing this proposal.

Real photographs: predictions and misses

The same model and development protocol produce both partial success and complete failure.

AEBIS_329 - useful localization, incomplete coverage

Input photograph



Saved prediction



Reference damage



Figure 1. RGB-only model, seed 20260915. The orange mask is the saved prediction; teal is the dataset reference. Display colours are added only to make the original binary masks readable. [R, p. 47]

Individual image: IoU 56.43%, precision 97.16%, recall 57.37%. Most predicted pixels are correct, but the model misses a substantial part of the annotated damage.

AEBIS_486 - a complete localization failure

Input photograph



Saved prediction



Reference damage



Figure 2. Same model seed. The prediction mask is empty, so the middle image has no orange overlay. This case was selected after scoring to show a failure. [R, p. 48]

Individual image: IoU, precision and recall are all 0%. This example does not tell us how frequently the model fails.

THE AGGREGATE PRELIMINARY RESULT

Across the reused AEBIS development set, the retained real-only model reports 61.941 +/- 3.081% IoU over three seeds. The two figures above are selected individual cases, not that average and not a new untouched test. The reference masks are 2D annotations, not measured 3D damage. [R, p. 24]

HTML demonstration: choose either photograph, toggle prediction and reference overlays, and inspect a close-up. The viewer replays saved outputs; it does not run a model.

One nick, two physical meanings

The linked data distinguishes the material that disappeared from the surface that remains exposed.

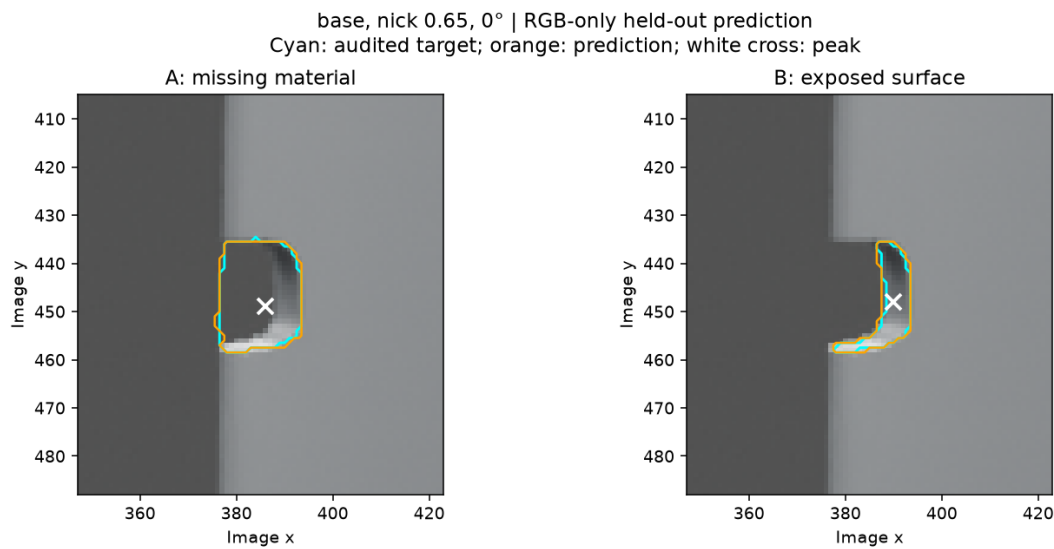


Figure 3. Actual saved two-output nick prediction. Left: missing-material footprint on the intact reference shape. Right: exposed cut surface on the damaged shape. Cyan = target, orange = prediction, white cross = score peak. [R, p. 46]

The reported run means are 82.90% IoU for the footprint and 70.21% for the exposed surface. They are **not individual-image scores** for Figure 3. The surface geometry is supplied; the model predicts image masks.

Does extra 3D training improve the mask?

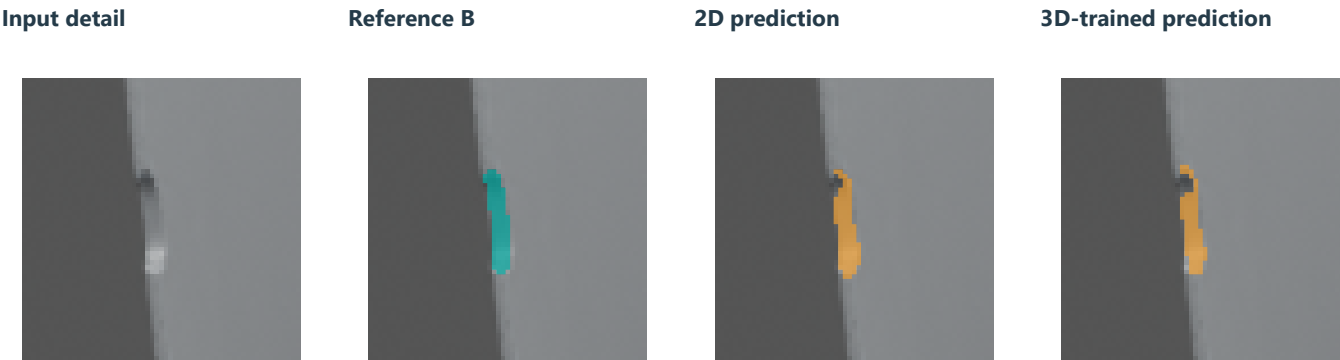


Figure 4. Shared display crop around the reference defect in p5_s1p2_v000. The same crop is shown for RGB, reference B and both predictions. Crop selection uses the reference only for display, not inference. [R, p. 49]

2D-only IoU: 75.76%. With 3D training: 68.04%. Both predicted peaks have zero target distance. The 3D-trained mask is worse in this saved case even though the point is correct.

WHY THIS BELONGS IN THE PROPOSAL

This is the exact distinction the proposed study must test: does extra geometry improve useful real-image masks? The example uses an initially unused nick position on one source CAD, not an unfamiliar blade design. Both models use RGB alone at prediction.

From an image mark to a checked 3D point

Existing demonstrations connect image evidence to known synthetic geometry.

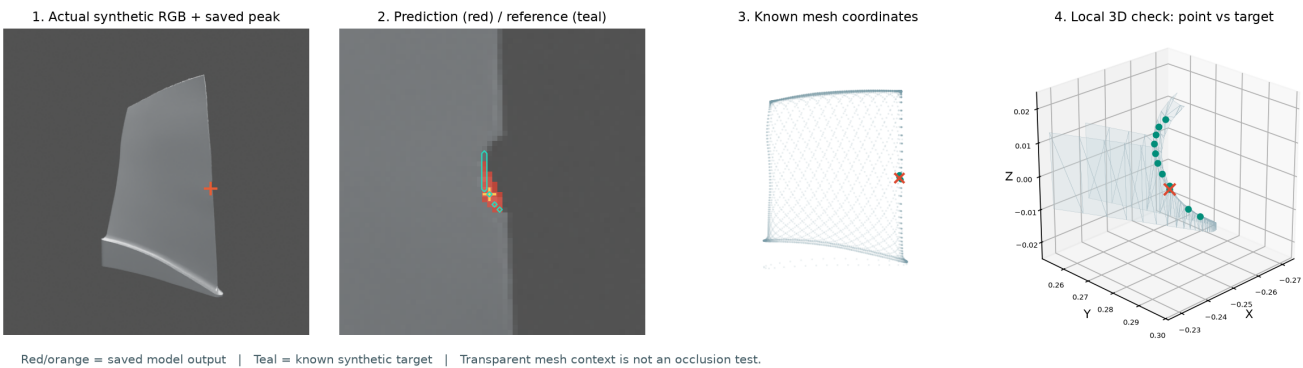


Figure 5. Saved image peak, local mask comparison, known mesh and local 3D check. Orange/red shows saved model output; teal shows the known synthetic target. The peak reaches the exposed surface, while mask IoU is only 20%. [V, p. 1]

Supplied: synthetic damaged mesh, camera and surface associations. **Predicted:** image mask and peak. The displayed mesh is reference geometry; it is not reconstructed from the photograph.

p1_s0p9 · source: damage_surface · 3D error: 0.000750 normalized units

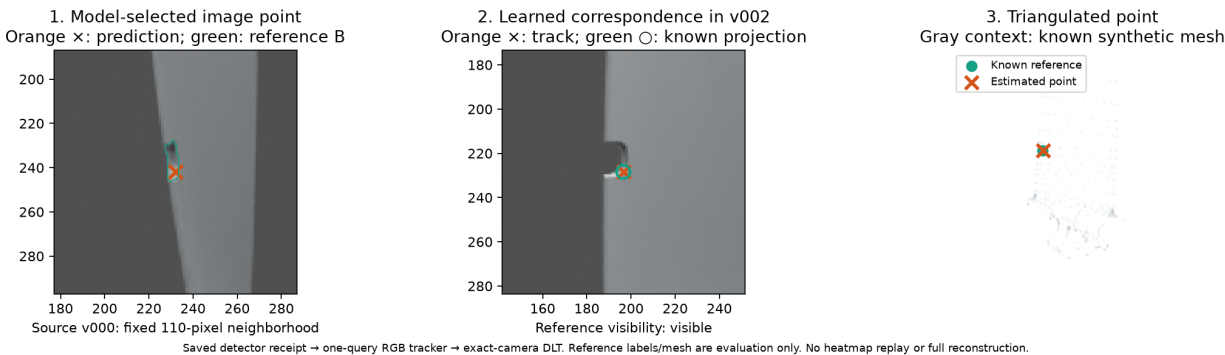


Figure 6. A detector-selected image point is tracked across generated views and triangulated with supplied exact cameras. The endpoint is checked against the known synthetic surface. [R, p. 52]

Selected case p1_s0p9: about 0.2180-pixel matching error and 0.0007496 normalized endpoint error. This is a useful point example, but the broader surface benchmark failed.

WHAT THE HTML DEMO SHOWS

Browse four saved 2D-to-3D outcomes: a correct point, a wrong surface point, an unmapped background point, and a removal-footprint case. The figures preserve those outcomes. No rotatable predicted mesh or new 3D inference is implied.

Figure 5 attribution: source geometry Safran / PLAID Rotor37; derived figure reproduced from the Main Jet visual evidence companion, which identifies its derived figures as CC BY-SA 4.0. [V, p. 5]

A visible dent-fitting result

Selecting a better prepared candidate can improve a controlled synthetic fit.

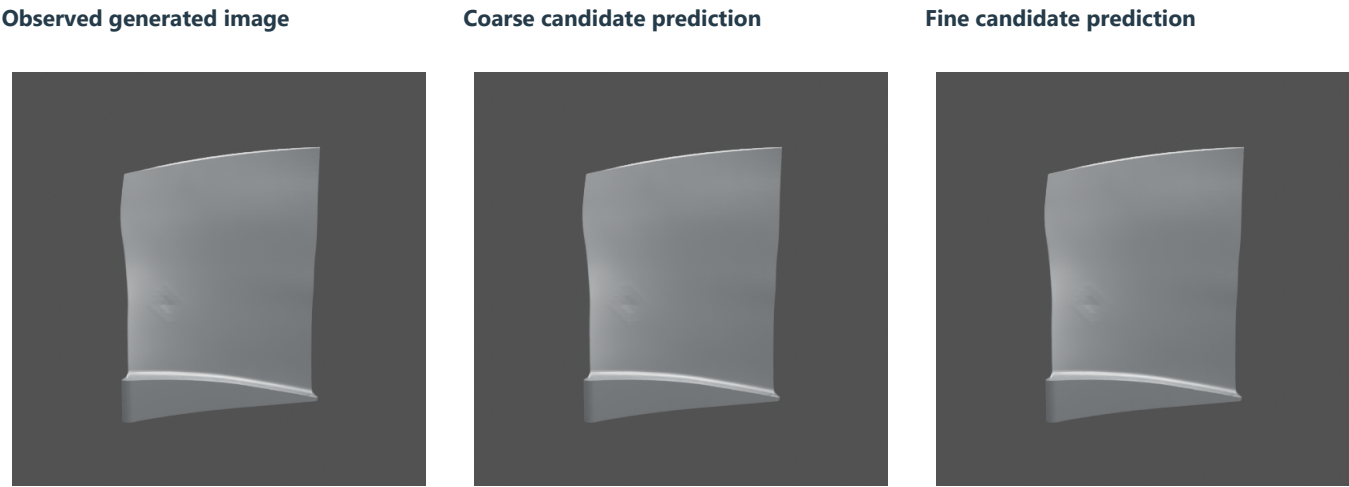
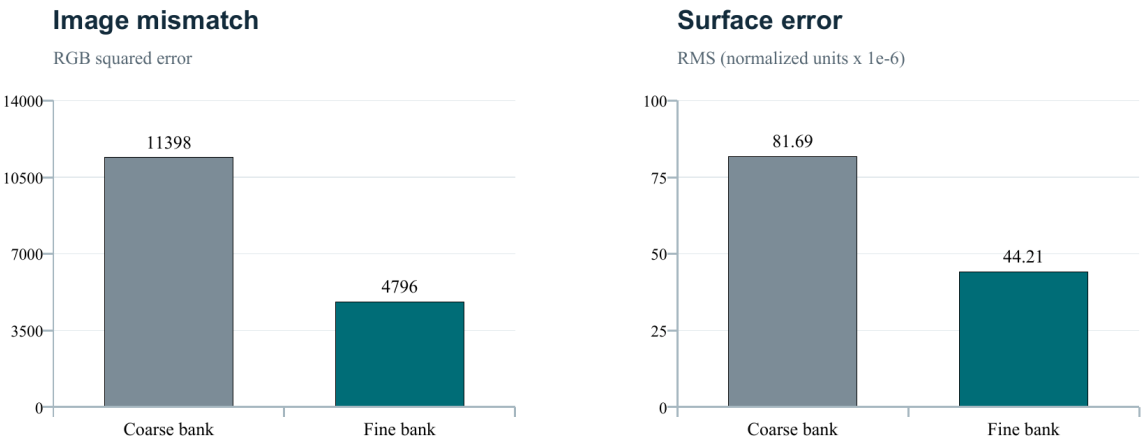


Figure 7. Actual saved images for obs_a. Both searches receive the healthy CAD, candidate family, possible centres and acquisition settings. These are nonlearned fits of a previously exposed synthetic case. [R, p. 60]



Same saved case: obs_a. Lower is better.

Figure 8. Replotted measurements for this same case, with zero-based axes. Lower is better. Image error and 3D surface error use different units and are not interchangeable. [R, p. 60]

Saved result: RGB squared error falls from 11,398 to 4,796; 3D RMS falls from 0.0000816854 to 0.0000442051 normalized units on the same 139-vertex population.

WHAT THIS SUPPORTS

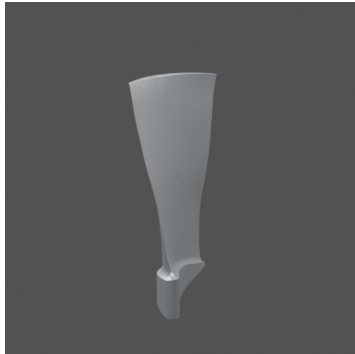
The image can distinguish candidates within a supplied family. This result supports a controlled fitting baseline; it does not establish a learned model, arbitrary shape recovery, real millimetre accuracy or transfer to new blade designs.

Why retain it? These outputs show what a constrained physical explanation can achieve, and provide a reference against which a future learned damage-recovery method would have to improve.

Learning the dent: progress and a failure

The learned field improves the small damaged region while moving too much healthy surface.

Actual input, source 5 / v004



Supplied inward/rim labels

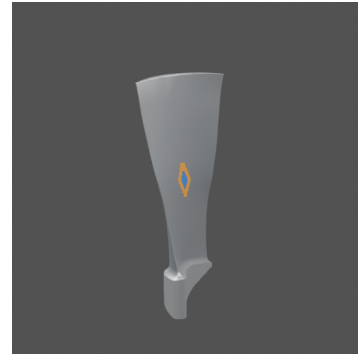


Figure 9. One displayed input view and its synthetic labels. The continuous-field regression actually receives seven healthy/observed RGB pairs, the healthy mesh and correspondences. [R, pp. 53, 102-104]

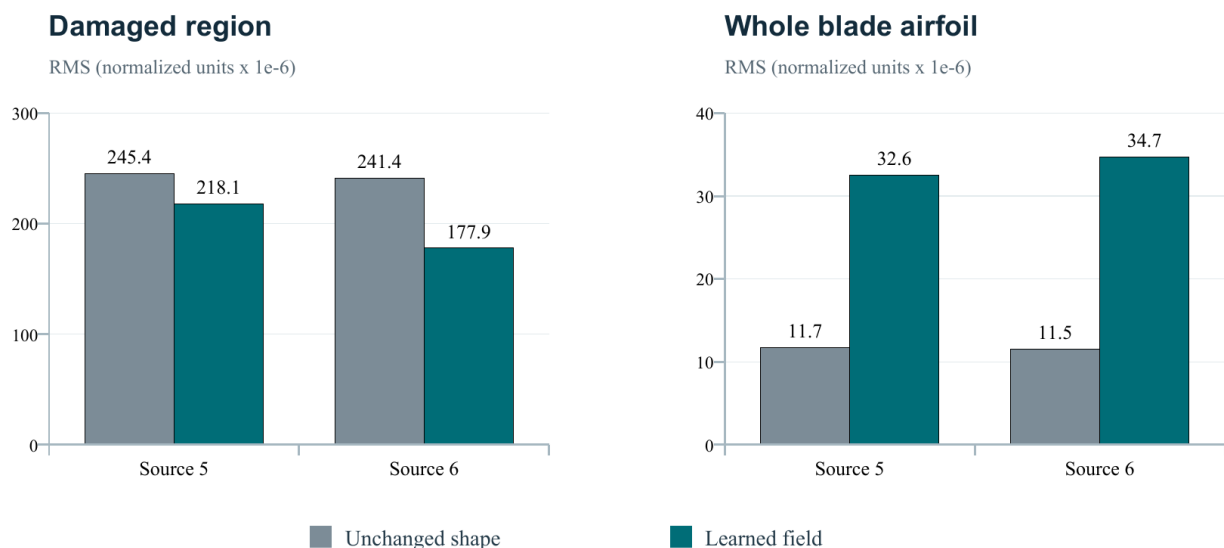


Figure 10. Reported continuous-field results on two exposed development shapes. Lower is better. Each panel has its own vertical scale; distances are normalized source units, not millimetres. [R, p. 103]

Damaged region: error improves by about 11% on source 5 and 26% on source 6. **Whole airfoil:** error becomes about 2.8x and 3.0x the unchanged-shape baseline. Around 48-49% of vertices in the healthy controls falsely move above the reported threshold. [R, p. 103]

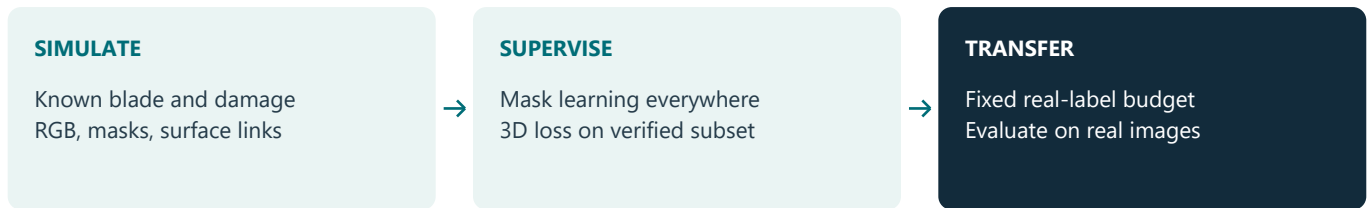
WHAT WAS ACTUALLY LEARNED

Six deterministic ridge-regression fits used only three nonzero training shapes. This is a continuous XYZ displacement prediction, not a finite-template search. However, the supplied snapshot contains no predicted-mesh PNG for this run, so no such picture is fabricated here. [R, pp. 102-104]

Research consequence: retain the result as preliminary evidence and a failure analysis. Reliable unknown 3D damage recovery remains a separate milestone; it is not required to complete the proposed RGB segmentation pilot.

One RGB model, extra teaching information

A photograph is the only input required when the proposed model is used.



1. Define the same target

The primary target is the visible defect region in a real image. Write an annotation guide with examples. A nick's missing-material footprint, its newly exposed cut surface, and a dent's inward region and raised rim are distinct concepts. Preserve these synthetic labels separately; do not silently treat them as interchangeable real masks.

2. Reuse the broad synthetic collection

Train with the eligible static RGB/mask data for appearance diversity. Give every synthetic-training condition the same broad images and the same eligible linked subset. Only the availability of the auxiliary geometric loss changes in the key comparison.

3. Reuse the existing geometric objective first

Start with the current 3D point-distance loss. Validated links measure location errors on the label-appropriate surface: intact geometry for the removal footprint, damaged geometry for the exposed cut surface. Keep identical mask and point heads in matched conditions; use zero geometric loss weight for the 2D baseline. Verify the implementation before freezing the protocol.

4. Adapt using exactly the same real labels

Fine-tune each matched model on the same real training images, masks and schedule. The real images require defect masks but do not require CAD models, measured depth, a healthy reference photograph or camera calibration for this primary task.

5. Test the transfer claim

Evaluate frozen predictions on the same real population, including appropriate healthy controls. Measure mask quality and false alarms. Keep synthetic 3D point metrics as secondary diagnostics, rather than substituting them for real-image success.

TRAINING RULE IN PLAIN LANGUAGE

Every qualified image teaches the model where damaged pixels are. A verified linked synthetic image can additionally teach where the damage lies on the simulated surface. Ordinary static synthetic images do not receive invented 3D labels. Geometry is extra training supervision; the model still uses RGB alone at deployment.

Scope of the first pilot

Begin the geometric comparison with nicks and dents, for which linked examples already exist. Other categories can supply ordinary 2D supervision when their mask definitions are compatible. A geometric benefit for scratches or corrosion cannot be claimed until their linked labels are validated. If label meanings cannot be reconciled, retain source-specific evaluations rather than reporting a misleading pooled score.

Later extension: prediction of unknown physical displacement or removed material would need separate experiments with suitable 3D truth, robust alignment and meaningful scale. It is outside the primary success criterion here.

Separate the effects of data and geometry

The key contrast uses identical synthetic images, model capacity and real-label access.

Condition	Training	Question answered
R - Real only	Fixed pretrained backbone, then the selected real training subset.	What does a strong model achieve without synthetic task training?
R+ - More real updates	Use the same selected real subset for extra updates matching the synthetic stage; then the same fine-tuning schedule.	Could extra optimization alone explain a synthetic-pretraining gain?
S - Synthetic + real	Static and linked synthetic RGB with 2D labels; then the same real subset.	Does the synthetic stage help at this real-label budget?
G - Geometry + real	Exactly S's synthetic images, plus the verified 3D loss on linked samples; then the same real subset.	Does geometry add value beyond the synthetic images?
D - 2D distance + real	Exactly S's images and training duration, with an auxiliary image-space distance loss; then the same real subset.	Is an apparent benefit specific to 3D information or explained by ordinary spatial supervision?

Fixed label budgets and run count

Proposed pilot budgets are 25 labelled real training images and the full eligible AEBIS training pool, currently 397 images. Reconfirm eligibility after the grouping audit, publish the exact image IDs and number of represented groups, and use a nested small subset. All conditions receive the same subset for a given budget. These numbers are planned settings, not results.

Use three paired random seeds: five conditions x two budgets x three seeds = 30 fine-tuned models. The three synthetic conditions need nine synthetic-stage checkpoints, shared across the real budgets. Evaluate these before real fine-tuning to report synthetic-only transfer without extra training. This is the planned core run cap; additional retries require a documented technical reason.

Controls to freeze before training

Use the same backbone, starting pretrained weights, resolution, augmentations, real sampling and fine-tuning length. Match synthetic image exposure, training duration and checkpoint selection across S, G and D. Give each condition an equal development-tuning allowance, then freeze settings. Record changes and failed runs. Select thresholds using development data only.

R+ matches total updates, but elapsed time can still differ by loss and data source. Report measured compute as well as performance. Compare synthetic training against both R and R+ when interpreting the value of the synthetic stage.

DATA USE MUST BE VISIBLE

Save each run's intended image list, the unique image IDs actually loaded, sampling counts and completion status. Report static-synthetic, linked-synthetic and real usage separately. Include varied successes, typical cases, failures and healthy examples so repeated demonstration images do not stand in for dataset coverage.

What would count as a useful result?

A point-location gain is supporting evidence; the primary outcome is a better real defect mask.

Evaluation populations

Use the repeatedly viewed 131 AEBIS images for development. Preserve the reported exposure status of existing AeBAD tests. They can support exploratory comparisons, but they cannot become fresh tests by renaming or reshuffling them. Obtain a genuinely unused, appropriately labelled real evaluation population for confirmatory claims; verify independence and prior exposure before selecting it. The 59 reserved AEBIS images require a freshness audit. [R, pp. 6, 36]

Keep known physical blades, capture sessions and video sequences separate across partitions. When blade identity is unavailable, disclose this and use documented capture or similarity groups as imperfect safeguards. New-blade-instance claims require independent physical blades; new-design claims require held-out blade designs. More views of the same surface establish neither.

Measure	Definition and reporting rule
Primary: real mask IoU	Intersection over union of predicted and annotated damaged pixels. Report the mean over damaged images, with a fixed aggregation rule and source-specific results. Empty predictions on damaged images score zero.
Precision and recall	Report both at a development-selected threshold. State aggregation. A larger predicted region may improve recall while generating extra false alarms.
Healthy false positives	Report false-positive pixel area relative to the blade region, and the proportion of healthy images triggering an alarm at a fixed alarm rule. Assess on eligible healthy data; AEBIS's 528-image pool alone cannot supply this.
Robustness and cost	Break down eligible categories, defect sizes and sources. Report training time, prediction time and real annotation count. Analyse RGB-only operation separately from supplied-geometry demonstrations.
Uncertainty	Report paired seed results. Where there are enough independent groups, use paired resampling of blade/capture groups for confidence intervals. Multiple seeds or video frames do not replace independent blades.

Proposed pilot decision rules

Advance the geometric contribution if G improves real IoU over S by at least 3 percentage points at the small-label budget, also improves over D, and does not increase healthy alarm rate by more than 2 percentage points at the predeclared operating rule. These are provisional planning thresholds to freeze before running, not observed effects or guarantees of statistical power.

Claim reduced label needs only if a lower-budget condition is within a predeclared 2 IoU percentage points of the full-budget real-only model while satisfying the same false-alarm constraint. Report the uncertainty; two budgets establish only this comparison, not a complete learning curve.

IF THE COMPARISON IS NEGATIVE OR INCONCLUSIVE

Report the result and retain the stronger baseline. If S helps but G does not, pursue the supported synthetic-data result without claiming a 3D benefit. If fresh data or sufficient independent groups are unavailable, label the study exploratory. A strong conference claim requires broader confirmation, not merely passing these pilot thresholds.

A bounded four-week pilot

Proposed working days, counted from the start of the study. Fresh data acquisition and annotation may extend the schedule.

Day	Planned work and reviewable output
1	Freeze the primary task, conditions, endpoints and scope; write a one-page protocol.
2	Reconcile source manifests, duplicates, label types and prior test exposure.
3	Audit blade/capture groups and label compatibility; list unresolved identities.
4	Review annotation examples; measure time per defect mask and identify fresh-data access.
5	Freeze eligible splits, real-label subsets and synthetic lists. Gate A: data readiness.
6	Replay a retained real baseline on development data; verify evaluation calculations.
7	Verify geometric and 2D-distance losses, valid links, masking and units.
8	Run a short technical pilot to measure memory, run time and sampling coverage.
9	Complete allowed development tuning; freeze settings, threshold-selection rules and run budgets.
10	Launch the synthetic-stage comparisons and save immutable run manifests.
11	Inspect run completion and technical failures; log repairs without selecting favourable cases.
12	Evaluate synthetic-only checkpoints using the declared development procedure.
13	Complete the small-real-budget comparisons with the paired seeds.
14	Complete the full-real-budget comparisons using the shared synthetic checkpoints.
15	Check training coverage and matched settings. Gate B: validity of the comparison.
16	Set thresholds on development predictions; freeze predictions and code before opening confirmatory labels.
17	Evaluate eligible real populations; compute paired differences and group uncertainty.
18	Audit healthy false alarms, small defects, source shifts and representative failures.
19	Package reproducible results, a readable report and an RGB inspection prototype.
20	Supervisor review: continue, revise the claim, or stop the unsupported mechanism.

SCHEDULE CONDITIONS

This is a planning estimate, not a promised completion date. It assumes usable existing code, available compute and ready labels. Re-estimate run duration after day 8. If fresh annotations are delayed, complete the development pilot and explicitly defer confirmation; do not reuse development results as a substitute.

Milestones: Gate A produces a defensible dataset and split; Gate B produces a valid matched comparison; the final review produces a documented decision and a reusable research package. Scientific failure to improve remains a valid outcome and should not trigger endless reruns.

A practical funding and supervision case

Resources should first support trustworthy labels and comparisons, then scale a supported result.

Requested support and budget basis

Resource	Purpose and how to estimate cost
Research time	One researcher for a proposed four-week pilot: curation, protocol verification, experiments and analysis. Publication development requires additional time.
Annotation and domain review	An annotator and blade-inspection reviewer to check masks and healthy examples. Estimate hours from a timed batch: images x minutes per image / 60, plus review.
GPU and rendering access	Nine shared synthetic-stage runs and 30 fine-tuning runs, including extra real-only update controls. Measure run hours and memory before costing.
Storage and reproducibility	Versioned manifests, labels, settings, checkpoints and predictions. Measure archive size and retention needs; preserve provenance and redistribution permissions.
Fresh evaluation data	Independently grouped real defect and healthy images with checked masks. The primary RGB task does not require matching CAD or a physical scanner.

Budget status: a monetary total awaits compute and staff rates, annotation volume and fresh-data access. Price the request after the timed pilot using institutional rates.

Risks and mitigations

Risk	Planned response
Insufficient independent real data	Audit exposure and capture groups early. Secure new annotations or limit the output to an exploratory study.
Synthetic appearance does not transfer	Retain the real-only baseline, analyse failure conditions and avoid expanding generation without a tested reason.
3D loss helps points but hurts masks	Use mask IoU and healthy alarms as the decision criteria; compare a matched 2D spatial loss.
Small shape diversity or incompatible labels	Report independent designs separately from image counts; defer unfamiliar-design and cross-category claims until supported.

Deliverables and value

Deliver a qualified manifest and annotation guide; matched experiments with image-usage records; a report with successes and failures; and an RGB prototype displaying defect masks. Release eligible data extensions and evaluation code with provenance. The practical goal is evidence about reduced real labelling effort, not a claim of operational maintenance readiness.

CONFERENCE AND GRANT POSITIONING

A strong submission needs differentiation from prior art, independent real evaluation and convincing controls. The near-term funding outcome is a reproducible feasibility result. Larger data collection or physical damage recovery can form a later, separately scoped work package.

Sources, definitions and claim boundaries

The proposal distinguishes the report's observations from its own planned experiments.

Primary sources

[R] **Main Jet Full Status Report.** User-supplied *MAIN_JET_FULL_STATUS_REPORT.pdf*, 105 pages, 22 September 2026. Inventory: pp. 5-10; principal training results: pp. 23-30; test exposure: p. 36; proposed directions in the report: p. 78; late learned-field update: pp. 102-105. This is the evidence source for project-specific counts and results; no access to the Linux experiment repository was used to independently reproduce them here.

[V] **Main Jet Visual Evidence.** User-supplied *MAIN_JET_VISUAL_EVIDENCE.pdf*, 5 pages, 21 September 2026. Saved synthetic image-to-surface examples; attribution and reproducibility on p. 5. Figures here are extracted from the source PDFs, not newly generated research outputs.

[1] **Elia Bonetto and Aamir Ahmad.** *ZebraPose: Zebra Detection and Pose Estimation using only Synthetic Data*. WACV 2026, pp. 6611-6620. [Official CVF publication](#); [author manuscript](#).

[2] **M. A. Mohammed Eltoum et al.** *BladeSynth: A High-Quality Rendering-Based Synthetic Dataset for Aero Engine Blade Defect Inspection*. Scientific Data 12, 1268 (2025). DOI: 10.1038/s41597-025-05563-y. [Full article](#).

[3] **Ankan Bhunia, Changjian Li and Hakan Bilen.** *Looking 3D: Anomaly Detection with 2D-3D Alignment*. CVPR 2024, pp. 17263-17272. [Author manuscript](#).

[4] **Shun-Hsiang Hsu and Mani Golparvar-Fard.** *Beyond Cracks: Synthetic Image and Geometry Generation for Computer Vision Detection and Severity Assessment of Diverse Concrete Surface Defects*. DefectSynth author project, listed as under review at Automation in Construction, 2026. [Author project and manuscript link](#). Publication status checked 22 September 2026; not treated as an accepted paper.

Plain-language glossary

Term	Meaning in this proposal
Synthetic image / render	A picture made by a simulator rather than a camera photographing a real blade.
Mesh / CAD / geometry	A digital 3D shape. A mesh describes its surface using points joined into small faces.
Mask / segmentation	A coloured-in region identifying the pixels that belong to damage.
Supervision / loss	The correct answer used during training / the penalty for a wrong prediction.
Development / test	Examples used to choose settings / examples reserved for final assessment. A repeatedly inspected test has lost that fresh-assessment role.
Generalization / label efficiency	Working on genuinely new examples / achieving a given result with fewer manually labelled training images.

DOCUMENT STATUS

Discussion proposal, version 1.1, 22 September 2026. It is neither a completed experiment report nor a funding award request with an approved budget. All schedules, new comparisons and numerical decision margins are proposed. Dataset and historical result counts are attributed to the supplied report.